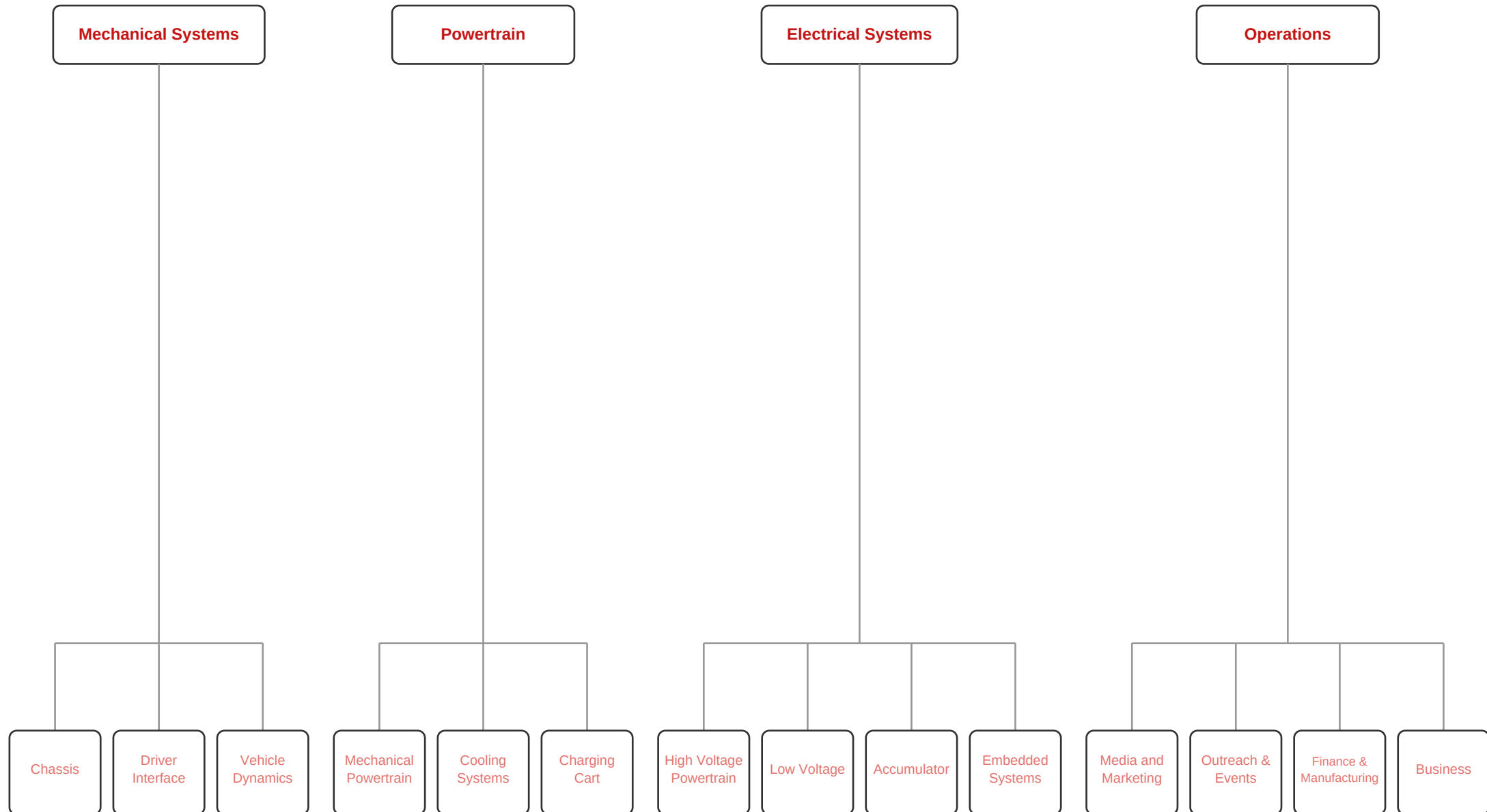


# CSG Racing: Team Structure

Formula Student EV Racing Team, City St George's, University of London



# Departments & Sub-Teams

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## Mechanical Systems

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Mechanical Systems designs, builds, and refines the physical machine that turns electrons into lap times: everything the driver sits in, sits on, and steers around a track. If you love CAD, composites, and seeing your parts bolted onto a real race car, this is where you belong.

### Chassis

Chassis designs and manufactures the car's structural backbone: the space-frame or monocoque that holds every other system together and keeps the driver safe in a crash. It's a hands-on mix of CAD, structural analysis, welding, and composite layup.

### Driver Interface

Driver Interface covers the harness, seat, and pedal box, including the brake and accelerator pedals: everything that connects the driver to the car.

### Vehicle Dynamics

Vehicle Dynamics covers the suspension system, from the unsprung mass through to the tyre-to-upright assembly, shaping how the car grips and handles the track.

## Powertrain

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Powertrain turns electrical energy into raw speed, engineering the mechanical powertrain and cooling systems that get power from the battery to the tarmac, plus the team's own mobile charging cart. Join if you want to work on the mechanical and electromechanical heart of the car's performance.

### Mechanical Powertrain

Mechanical Powertrain team is responsible for mounting and connecting the motor, differential and drive system.

### Cooling Systems

Cooling Systems team is responsible for designing and integrating the cooling system to keep the motor and inverter within safe operating temperatures.

### Charging Cart

Charging Cart team is responsible for the safe transport and charging of the accumulator including the cart braking system and charging connections.

## Electrical Systems

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Electrical Systems keeps the car's power and control network alive, from the battery pack that stores every joule to the embedded brains that keep everything talking to each other safely. If you love the idea of your circuit board deciding whether the car drives or shuts down, this is your department.

*These are the department's core sub-teams, but placement isn't fixed. If another sub-team needs an extra pair of hands for a smaller task, you may be asked to help out there too.*

## High Voltage Powertrain

High Voltage Powertrain team is responsible for the electrical side of the HV drivetrain, including the inverter, discharge circuit, instrument panel, HV cabling and motor controller setup.

## Low Voltage

Low Voltage is responsible for wire harness design, the safety and shutdown circuits (SDC), low-voltage power control systems, and general circuit design across the car.

## Accumulator

Accumulator is responsible for the tractive system accumulator container (TSAC) and the accumulator itself, built and certified to strict safety standards.

## Embedded Systems

Embedded Systems is in charge of the car's firmware, sensor code, and ECU communication (CAN), along with data acquisition, logging live sensor and system data so the team can review performance after every run.

# Operations

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Operations is the engine room behind the engine room, handling the sponsorship, budgeting, communications, and events that keep the whole team funded, visible, and running smoothly. It's the perfect fit if you want to shape the team's success without picking up a spanner.

## Media and Marketing

Media and Marketing is responsible for posts across every platform, handling photography and video editing to spread the word about the team.

## Outreach & Events

Outreach & Events contacts sponsors for events and handles general communications, planning every kind of event the team runs.

## Finance & Manufacturing

Finance works with every sub-team to complete the cost and manufacturing documentation required for competition, tracking and documenting all costs incurred.

## Business

Business is responsible for the business plan proposal, one of Formula Student's static events where teams pitch the commercial case for their car to a panel of industry judges at Silverstone. Learn more on the [IMechE Formula Student website](#).